Knowledge Discovery
Comprehensive Unique Questions Bank

Compiled Question Bank

1 True/False Questions

1. Feature scaling is the process of converting data to a uniform scale. [True]
2. The Euclidean distance is a measure of similarity between two data points. [False]
3. Data reduction techniques involve compressing and summarizing large amounts of data. [True]
4. When min Sup increase, frequent itemset increase. [False]
5. Generating association rules from frequent itemset is the difficult part of mining operation. [False]
6. Data preprocessing is an optional step in the data mining process. [False]
7. Every subset of a frequent itemset is also frequent. [True]
8. The support of every subset J of I is at least equal to that of the support of itemset. [True]
9. The confidence of a rule is the percentage of transactions that contain the antecedent of the rule that also contain the consequent. [True]
10. minSup a sufficient strength in terms of conditional probabilities. [False]
11. The basic idea of Apriori algorithm is to use the downward closure property to prune the candidate search space. [True]
12. Discretization is the process of transforming continuous data into discrete values. [True]
13. One of the key challenges in data mining is dealing with the curse of dimensionality. [True]
14. Data integration is the process of combining data from different sources into a single dataset. [True]
15. The Manhattan distance is equivalent to the L1 norm. [True]
16. The Apriori algorithm is a bottom-up approach to association rule mining that starts with frequent itemsets and builds larger itemsets by combining them. [True]
17. Association rule mining is used to identify patterns in transactional data. [True]

2 Multiple Choice Questions

18. Which of the following is a common method for reducing the number of association rules generated by a frequent pattern mining algorithm?

    A. Increasing the minimum support threshold
    B. Decreasing the minimum support threshold
    C. **Increasing the minimum confidence threshold**
    D. Decreasing the minimum confidence threshold

19. Which of the following is a common approach for handling noise and outliers in frequent pattern mining?

    A. Increasing the minimum support threshold
    B. Decreasing the minimum support threshold
    C. **Removing transactions with low support**
    D. Using clustering to identify groups of similar transactions

20. What is the relationship between data mining and knowledge discovery in databases (KDD)?

    A. Data mining is a synonym for KDD.
    B. KDD is a synonym for data mining.
    C. **Data mining is a step in the KDD process.**
    D. KDD is a step in the data mining process.

21. Which of the following is a method for outlier detection?

    A. **Local Outlier Factor (LOF)**
    B. Principal Component Analysis (PCA)
    C. Linear Discriminant Analysis (LDA)
    D. Support Vector Machines (SVM)

22. What is the primary goal of data reduction in data preprocessing?

    A. **To eliminate irrelevant or redundant data**
    B. To combine multiple datasets into a single dataset
    C. To convert unstructured data into structured data
    D. To scale data to a standardized range

23. What is the primary challenge of data preprocessing in data mining?

    A. Finding the right data
    B. Handling big data
    C. **Cleaning and integrating data from multiple sources**
    D. Building accurate models

24. What is the primary difference between data mining and machine learning?
    - A. Data mining is a subset of machine learning.
    - B. Machine learning is a subset of data mining.
    - C. **Data mining focuses on discovering patterns and relationships in data, while machine learning focuses on making predictions and decisions.**
    - D. Machine learning focuses on discovering patterns and relationships in data, while data mining focuses on making predictions and decisions.
25. What is the primary goal of data discretization in data preprocessing?
    - A. To correct errors and inconsistencies in the data
    - B. To reduce the size of the data
    - C. To transform the data into a more meaningful format
    - D. **To convert continuous data into categorical data**
26. Which of the following is a disadvantage of using the Euclidean distance for similarity measurement?
    - A. It is not suitable for high-dimensional data
    - B. **It is sensitive to outliers**
    - C. It is not suitable for continuous data
    - D. It assumes that the data is normally distributed
27. What is the primary goal of data normalization in data preprocessing?
    - A. To correct errors and inconsistencies in the data
    - B. To reduce the size of the data
    - C. To transform the data into a more meaningful format
    - D. **To scale the data to a common range**
28. Which of the following is an example of a sequential pattern mining problem?
    - A. Finding frequent itemsets in a transaction database
    - B. Identifying groups of customers with similar purchasing behavior
    - C. **Detecting patterns of activity in a time series of events**
    - D. Classifying images based on their content
29. Which of the following distance measures is not appropriate for categorical data?
    - A. **Euclidean distance**
    - B. Hamming distance
    - C. Jaccard distance
    - D. Minkowski distance

Considering the following transactions table, answer questions 30, 31, and 32:

TID	Set of Items
1	{Bread, Butter, Milk}
2	{Eggs, Milk, Yogurt}
3	{Bread, Cheese, Eggs, Milk}
4	{Eggs, Milk, Yogurt}
5	{Cheese, Milk, Yogurt}

30. Support of {Cheese, Milk} is:

    A. 0.2

    B. 0.4

    C. 0.6

    D. 3

31. For minSupport = 0.5, is {Eggs, Milk} considered a frequent itemset?

    A. Yes

    B. No

    C. No available data to judge

    D. It depends on minimum confidence value

32. For minSupport = 0.3, Which of the following itemsets is a maximal frequent itemset?

    A. {Bread, Milk}

    B. {Cheese, Milk}

    C. {Eggs, Milk, Yogurt}

    D. All of the Above

33. Which of the following is a disadvantage of association rule mining?

    A. It can find interesting patterns that might not be discovered using other techniques

    B. It can be computationally expensive for large datasets

    C. It can only find patterns with a minimum support threshold

    D. It cannot handle categorical data.

34. Which of the following is not a goal of data mining?

    A. Predictive modeling

    B. Anomaly detection

    C. Data warehousing

    D. Descriptive modeling

35. Which of the following is not a common data mining application?
    A. Fraud detection
    B. Recommender systems
    C. Web search engines
    D. **Spreadsheet software**
36. What is the primary difference between data mining and traditional statistics?
    A. **Data mining focuses on finding patterns and relationships, while traditional statistics focus on hypothesis testing and parameter estimation.**
    B. Data mining uses unsupervised learning, while traditional statistics use supervised learning.
    C. Data mining uses classification algorithms, while traditional statistics use regression analysis.
    D. There is no difference between data mining and traditional statistics.
37. Which of the following is not a typical data mining task in business applications?
    A. Market basket analysis
    B. Customer segmentation
    C. Fraud detection
    D. **Genetic sequencing**
38. Which of the following is not a common data preprocessing step in data mining?
    A. Data cleaning
    B. Data integration
    C. **Data warehousing**
    D. Data reduction
39. What does the support represent in Frequent Itemset Mining?
    A. Maximum value in the dataset
    B. Minimum value in the dataset
    C. **Frequency of an itemset in the dataset**
    D. Confidence level of a rule
40. In the context of hash trees, how does the maximum depth of the tree impact the efficiency of frequent itemset mining?
    A. It increases the computational complexity
    B. **It accelerates the counting process by reducing traversal**
    C. It limits the number of itemsets that can be stored
    D. It results in a higher number of false negatives

41. What is the target variable typically used for in data mining tasks?

    A. To manipulate the data
    B. To store large arrays and matrices
    C. To clean the data
    D. **To predict or classify based on input features**

42. How does a hash tree accelerate the counting of supports for candidate itemsets in frequent itemset mining algorithms?

    A. By organizing itemsets in a sequential manner
    B. By hashing each item in the transaction database independently
    C. **By examining transactions with a small number of relevant candidates**
    D. By including infrequent itemsets in the analysis for comparison

43. How does the support metric differ from the confidence metric in association rule analysis?

    A. Support measures item popularity, while confidence measures rule strength
    B. **Support determines the probability of items being purchased together, while confidence evaluates rule significance**
    C. Support provides insights into customer purchasing patterns, while confidence simplifies the interpretation of machine learning models
    D. Support calculates lift values, while confidence assesses the predictive power of rules

44. Which of the following is part of the Data Mining Process?

    A. Feature Extraction
    B. Data Cleaning and Integration
    C. Analytical Processing and Algorithms
    D. **All of the above**

45. How does the symmetric nature of a hash tree contribute to the accurate counting of supports?

    A. By ensuring that each itemset is equally represented in the tree
    B. By allowing for efficient mapping of items to specific nodes
    C. By enabling the identification of frequent patterns with high confidence
    D. **By reducing the occurrence of false associations between itemsets**

46. The confidence of a rule $X \Rightarrow Y$ is calculated as:

    A. $sup(X)/sup(Y)$
    B. $sup(Y)/sup(X)$
    C. $sup(X \cap Y)/sup(X)$
    D. $sup(X \cap Y)/sup(Y)$

47. What property states that when an itemset is contained in a transaction, all its subsets will also be contained in the transaction?

A. Support Monotonicity Property
B. **Downward Closure Property**
C. Confidence Property
D. Association Property

48. What is involved in handling incorrect and inconsistent entries during data cleaning?

    A. Reversing the entries
    B. **Replacing incorrect values with averages**
    C. Converting incorrect entries to positive values
    D. Ignoring the incorrect entries

49. Which factor does the Apriori algorithm address to enhance performance in association rule mining?

    A. Handling missing values in datasets
    B. **Reducing the search space for better performance**
    C. Detecting outliers in data clusters
    D. Optimizing the visualization of rule networks

50. Which of the following is a primary goal of using association rule measures?

    A. Minimizing the support threshold
    B. Maximizing the lift value
    C. **Extracting meaningful relationships from data**
    D. Decreasing the confidence level of rules

51. When integrating data from multiple sources, what is the main goal?

    A. To eliminate all existing entries
    B. **To combine data into one cohesive dataset**
    C. To separate data into individual categories
    D. To compare data sources and select the best one

52. What is the primary focus of data mining algorithms?

    A. Data visualization
    B. **Data prediction and modeling**
    C. Data storage
    D. Data collection

53. In a hash tree, what does each internal node typically represent?

    A. A lexicographically sorted itemset
    B. **A random hash function mapping an item to its parent**
    C. A list of infrequent itemsets
    D. A support value for a specific item